

Updating Asahi Linux Firmware on NixOS

<https://gemini.google.com/app/d1945fa2a50fcf07>

User prompt: Okay, here is the status. I am running Asahi Linux (7.1.13) on my m1pro machine. I wanted to update the latest firmware. So I upgraded the MacOSX to 27 and re-run the asahi linux installer and chose upgrade existing m1n1 and completed it. This caused my NixOS to enter a boot loop, which was caused by API change in macsmc. I fixed it by adding a manual module_blacklist in the uboot menu. So now I can boot into NixOS. I want to update my firmware (it is a private github repo and imported as a flake). But I don't see any new firmware files under /boot. Where can I find it? **➤**

```

ls /boot/asahi Permissions Size User Group Date Modified Name .rwxr-xr-x 692 root :
10 Oct 19:58 AdminUserRecoveryInfo.plist* .rwxr-xr-x 562 root : 10 Oct 19:58
stub_info.json* .rwxr-xr-x 70k root : 7 Jan 18:11 BuildManifest.plist* .rwxr-xr-x 358 root :
7 Jan 18:11 RestoreVersion.plist* .rwxr-xr-x 26M root : 7 Jan 18:11
kernelcache.release.mac13j* .rwxr-xr-x 606 root : 7 Jan 18:11 SystemVersion.plist* .rwxr-xr-
x 29M root : 7 Jan 18:11 all_firmware.tar.gz* .rwxr-xr-x 228k root : 7 Jan 18:11
installer.log* ~ ➤ ls /boot Permissions Size User Group Date Modified Name drwxr-xr-x - root :
10 Oct 11:33 loader/ drwxr-xr-x - root : 10 Oct 06:41 m1n1/ drwxr-xr-x - root : 7
Jan 18:11 .Spotlight-V100/ drwxr-xr-x - root : 7 Jan 18:11 .Trashes/ drwxr-xr-x - root :
7 Jan 18:11 .fsevents/ drwxr-xr-x - root : 7 Jan 18:11 vendorfw/ drwxr-xr-x - root :
7 Jan 18:11 asahi/ drwxr-xr-x - root : 7 Jan 13:26 EFI/ .rwxr-xr-x 728 root : 1 Jan
2000 ubootefi.var* ~ ➤ ls /boot/vendorfw Permissions Size User Group Date Modified Name
drwxr-xr-x - root : 7 Jan 18:11 u-boot/ .rwxr-xr-x 32M root : 7 Jan 18:11
firmware.cpio* .rwxr-xr-x 38k root : 7 Jan 18:11 manifest.txt* .rwxr-xr-x 33M root : 7 Jan
18:11 firmware.tar* ~ ➤ ls /boot/EFI Permissions Size User Group Date Modified Name drwxr-
xr-x - root : 10 Oct 06:41 nixos/ drwxr-xr-x - root : 31 May 11:12 systemd/ drwxr-xr-
x - root : 31 May 11:12 BOOT/ drwxr-xr-x - root : 7 Jan 13:26 Linux/ ~ ➤ ls
/boot/EFI/nixos Permissions Size User Group Date Modified Name drwxr-xr-x - root : 10 Oct
06:41 .extra-files/ .rwxr-xr-x 26M root : 31 May 11:16
xny11ddr5x6935z51nvgw0yw9ppqyfvq-initrd-linux-fairydust-6.19.14-initrd.efi* .rwxr-xr-x 65M root
: 31 May 11:16 qm8saldmka8jjchx1v3llf1a0m10986p-linux-fairydust-6.19.14-Image.efi* ➤ ls
/boot/EFI/BOOT Permissions Size User Group Date Modified Name .rwxr-xr-x 150k root : 1
Jan 1980 BOOTAA64.EFI* ~ ➤ ls /boot/EFI/Linux ~ ➤ ls /boot/m1n1 Permissions Size User
Group Date Modified Name .rwxr-xr-x 7.6M root : 1 Jan 1980 boot.bin* ~ ➤ ls /boot/loader
Permissions Size User Group Date Modified Name drwxr-xr-x - root : 10 Oct 06:41 entries/
drwxr-xr-x - root : 7 Jan 13:26 keys/ .rwxr-xr-x 32 root : 10 Oct 11:33 random-seed*
.rwxr-xr-x 62 root : 10 Oct 06:41 loader.conf* .rwxr-xr-x 6 root : 7 Jan 13:26
entries.srel* ~ ➤ lsblk -o NAME,FSTYPE,SIZE,LABEL,MOUNTPPOINT NAME FSTYPE SIZE
LABEL MOUNTPPOINT mtddb0 1M mtddb1 16K mtddb2 448K
nvme0n1 931.8G |—nvme0n1p1 apfs 500M |—nvme0n1p2 apfs 186.3G |—nvme0n1p3
apfs 2.3G |—nvme0n1p4 vfat 477M EFI - NIXOS /boot |—nvme0n1p5 ext4 737.3G nixos
/ |—nvme0n1p6 apfs 5G nvme0n2 3M nvme0n3 128M

```

Response: The reason you don't see updated firmware files under /boot is that the **"upgrade existing m1n1"** action in the Asahi installer only updates the bootloader payload (boot.bin / m1n1) inside your EFI system partition (nvme0n1p4). It deliberately **skips re-extracting vendor firmware** (all_firmware.tar.gz) from macOS to save time. Notice that only stub_info.json and AdminUserRecoveryInfo.plist show today's date (10 Oct), while all_firmware.tar.gz and /boot/vendorfw are still dated 7 Jan.

Your macOS APFS containers (nvme0n1p1, p2, p3, p6) hold the newly installed macOS system files. You can extract the updated firmware in one of two ways:

Option 1: Extract Directly from NixOS (Recommended)

Since your NVMe controller is functional in NixOS, you can mount your main macOS container volume using `apfs-fuse` and run `asahi-fwextract` directly against it.

1 Run `apfs-fuse` and extract firmware

Run via `nix-shell`

Launch an ephemeral shell containing `apfs-fuse` and the Asahi firmware extraction tooling:

```
nix-shell -p apfs-fuse asahi-firmware-extractor
```

Mount your main macOS volume (typically `nvme0n1p2`, the 186.3G APFS container) and run the extractor:

```
sudo mkdir -p /mnt/macos /tmp/new_fw
sudo apfs-fuse -o allow_other /dev/nvme0n1p2 /mnt/macos
sudo asahi-fwextract /mnt/macos /tmp/new_fw
```

Verification: Verify extraction succeeded by checking `ls -la /tmp/new_fw`. You should see generated `.tar` or `.cpio` firmware archives with current timestamps.

2 Unmount the APFS container

Cleanup

Unmount the macOS APFS volume once extraction is complete:

```
sudo umount /mnt/macos
```

Verification: Run `mount | grep macos` to verify the filesystem is cleanly unmounted.

3 Package and import into your Flake

Firmware repository update

Copy the extracted firmware files into your private firmware Git repository:

```
cp /tmp/new_fw/* /path/to/your/private-fw-repo/
```

Update your flake's lockfile (`nix flake update <fw-input>` or update the commit hash/SRI hash), rebuild your NixOS system, and verify that the new firmware is picked up by `apple-silicon-support / hardware.asahi.extractPeripheralFirmware`.

Option 2: Re-run Asahi Installer to Force Firmware Extraction

If you prefer using the official installer script:

1. Reboot into **macOS**.
2. Open Terminal and re-run the web installer:

```
curl -sL https://alx.sh | sh
```

3. Instead of choosing *Upgrade existing install*, look for options related to updating/extracting firmware for an existing OS install, or re-run the firmware export step. This will overwrite `/boot/asahi/all_firmware.tar.gz` and `/boot/vendorfw/firmware.tar` on the EFI partition with the new blobs.
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